METRIC

MIL-DTL-20330B
25 July 2020
SUPERSEDING
MIL-C-20330A
23 July 1962

DETAIL SPECIFICATION

CELLULOSE WOODPULP (SULFATE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This specification covers the requirements, examinations, and tests for cellulose woodpulp (sulfate) for use in nitrocellulose manufacturing.

1.2. Classification. Cellulose woodpulp (sulfate) is classified into the following classes:

Class 1: For combustible cartridge case (CCC) applications
Class 2: For solvent and solventless propellant applications
Class 3: For propellants made from lacquer applications

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2. Government documents.

2.2.1. Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1795	-	Standard Test Method for Intrinsic Viscosity of Cellulose

(Copies of this document are available from https://www.astm.org/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO-692	-	Pulps - Determination of Alkali Solubility

(Copies of these documents are available online at www.iso.org.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI T 211	-	Ash in wood, pulp, paper and paperboard: combustion at 525°C
TAPPI T 230	-	Viscosity of pulp (Capillary Viscometer Method)
TAPPI T 234	-	Coarseness of pulp fibers
TAPPI T 235	-	Alkali solubility of pulp at 25°C
TAPPI T 400	-	Sampling and accepting a single lot of paper, paperboard, containerboard, or related product, Standard Practice
TAPPI T 410	-	Grammage of paper and paperboard (weight per unit area)
TAPPI T 411	-	Thickness (Caliper) of Paper, Paperboard, and Combined Board
TAPPI T 412	-	Moisture in Pulp, Paper and Paperboard
TAPPI T 807	-	Bursting Strength of Paperboard and Linerboard

(Copies of these documents are available online at www.tappi.org.)

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2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. Required inspection.

3.1.1. First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2. Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2. Material. Cellulose supplied in accordance with this specification shall be made from softwood only.

3.3. Physical and chemical properties. Cellulose shall conform to the physical and chemical requirements specified in Table I.

TABLE I. Physical and chemical requirements for cellulose.

Requirement	Characteristics
Basis weight (g/m^2)	420 – 930 Class 3 683 – 850 Class 1,2
Thickness (mm)	0.7 – 2.0 Class 3 0.75 – 1.91 Class 1,2
Moisture (%)	10.0 maximum (max.)
Coarseness (mg/100 m)	Information only
Burst strength - Mullen (kPa)	550 – 1250 Class 1 and 2 only
Intrinsic viscosity (dL/g)	3.5 – 10.0 Class 3 5.6 min. Class 1,2
Viscosity (cP)	4.0 – 36.0 Class 3 9.0 min. Class 1,2
Alkali soluble matter (S10) (%)	9.0 max. Class 3 4.0 max. Class 1,2
Ash (%)	0.30 max.

3.4. Workmanship. Cellulose shall maintain a uniform composition, texture, and color; be free from visible impurities, tears, lumps, and foreign material such as dirt, wooden particles or any other foreign matter.

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4. VERIFICATIONS

TABLE II. Requirement/verification cross reference matrix.

Method of Verification 1- Analysis 2- Demonstration 3- Examination 4- Test		Classes of Verification A- First article B- Conformance						
Section 3 Requirement	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Material			X		X	X	4.4
3.3	Basis weight			X	X	X	X	4.5.1
3.3	Thickness			X	X	X	X	4.5.2
3.3	Moisture			X	X	X	X	4.5.3
3.3	Coarseness			X	X	X	X	4.5.4
3.3	Burst strength - Mullen			X	X	X	X	4.5.5
3.3	Intrinsic viscosity			X	X	X	X	4.5.6
3.3	Viscosity			X	X	X	X	4.5.7
3.3	Alkali soluble matter (S10)			X	X	X	X	4.5.8
3.3	Ash			X	X	X	X	4.5.9
3.4	Workmanship			X		X	X	4.6

4.1. Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2. First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table III. Samples from not less than five (5) rolls of cellulose collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

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TABLE III. First article acceptance tests.

Category	Examination or Test	Requirement Paragraph	Inspection Method
Critical	None Defined		
Major			
101	Material	3.2	4.4
102	Basis weight	3.3	4.5.1
103	Thickness	3.3	4.5.2
104	Moisture	3.3	4.5.3
105	Coarseness	3.3	4.5.4
106	Burst strength - Mullen	3.3	4.5.5
107	Intrinsic viscosity	3.3	4.5.6
108	Viscosity	3.3	4.5.7
109	Alkali soluble matter (S10)	3.3	4.5.8
110	Ash	3.3	4.5.9
Minor			
201	Workmanship	3.4	4.6

4.2.1. Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3. Conformance inspection.

4.3.1. Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each lot shall consist of a quantity of cellulose that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one or more acceptable batches of cellulose produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2. Sampling. Cellulose sampling shall be in accordance with TAPPI method T 400. Remove and discard at least one layer from the roll to reveal cellulose that has not been part of the outer layer. Cut a piece of the cellulose from each roll to be tested at not less than 0.60 meters, across the entire width of the roll. Place the sample into a large plastic bag and close it tightly.

4.3.3. Rejection. Cellulose shall be rejected in accordance with TAPPI method T 400.

4.3.4. Classification of characteristics. Sampling plan for major and minor defects shall be accordance with 4.3.4.1.

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4.3.4.1 Cellulose.				
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None Defined			
Major				
101	Material	4.4	3.2	4.4
102	Basis weight	4.3.2	3.3	4.5.1
103	Thickness	4.3.2	3.3	4.5.2
104	Moisture	4.3.2	3.3	4.5.3
105	Coarseness	4.3.2	3.3	4.5.4
106	Burst strength - Mullen	4.3.2	3.3	4.5.5
107	Intrinsic viscosity	4.3.2	3.3	4.5.6
108	Viscosity	4.3.2	3.3	4.5.7
109	Alkali soluble matter (S10)	4.3.2	3.3	4.5.8
110	Ash	4.3.2	3.3	4.5.9
Minor				
201	Workmanship	4.3.2	3.4	4.6

4.4. Material. Verify by examination of objective evidence that the cellulose is made from softwood (see 6.2.f).

4.5. Test methods and procedures. All tests shall be performed at ambient temperature, unless otherwise specified.

4.5.1. Basis weight. The basis weight shall be determined in accordance with TAPPI method T 410.

4.5.1.1. Basis weight alternate. Cut four 8 cm x 8 cm ± 2.0 mm squares from randomly selected areas of the cellulose sample using a standard paper cutter. Weigh each square on an analytical balance to the nearest 0.0001 g. Determine the average weight of the cellulose samples. Determine the grammage (weight per unit area in g/m^2) by the following formula:

$$\text{Grammage} = \frac{A}{(64 * 0.0001)}$$

Where:
A = average weight, g

4.5.2. Thickness. The thickness shall be determined in accordance with TAPPI method T 411.

4.5.2.1. Thickness alternate. Using the four 8 cm x 8 cm squares selected for the basis weight (see 4.5.1.1) determination, determine the thickness of each square (in mm) in four different locations using a standard handheld outside micrometer, and avoiding the edges of the sample. Determine the average thickness of the cellulose sample.

4.5.3. Moisture. The moisture shall be determined in accordance with TAPPI method

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T 412.

4.5.4. Coarseness. The soft wood coarseness shall be determined in accordance with TAPPI method T 234.

4.5.5. Burst strength. The burst strength test shall be determined in accordance with TAPPI method T 807.

4.5.6. Intrinsic viscosity. The intrinsic viscosity shall be determined in accordance with ASTM D1795.

4.5.7. Viscosity. The viscosity shall be determined in accordance with TAPPI method T 230 (0.5 percent Cupriethylenediamine Method). The viscometer shall be filled by immersing its small-diameter leg into the solution and drawing the liquid into the instrument by applying suction to the other end, filling the liquid level to the second etch mark. The volume of solution needed to fill to this point can be determined for each viscometer, and, for future samples, this volume may be added directly to large-diameter leg of the viscometer.

4.5.8. Alkali soluble matter (S10). The alkali soluble matter (S10) shall be determined in accordance with ISO method 692.

4.5.8.1. Alkali soluble matter (S10) alternate. The alkali soluble matter (S10) shall be determined in accordance with TAPPI method T 235.

4.5.9. Ash content. The ash content shall be determined in accordance with TAPPI method T 211.

4.6. Workmanship. Cellulose shall maintain a uniform composition, texture, and color; be free from visible impurities, tears, lumps, and foreign material such as dirt, wooden particles or any other foreign matter. Failure to meet the workmanship inspection during or following production processes shall result in lot rejection.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The intended use of the cellulose woodpulp (sulfate) is in the manufacture of nitrocellulose.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, grade and date of this specification
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
c. Provisions for submission/resubmission of first article samples
d. Quantity required and delivery schedules
e. Levels of preservation and packing (see 5.1)
f. Certificate(s) of Analysis reporting the species of softwood used and the test results for the rolls or lots tested.

6.3. Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, CCDC AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.4. Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, CCDC AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.5. Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.3) responsible for preparation of this specification.

6.6. Incorporated changes. ECP Y20Q2005 provides documentation for noted changes.

6.7. Government limited liability. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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6.8. Subject term (key word) listing.

Nitrocellulose
Propellant

6.9. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS

Preparing activity:
Army - AR
(Project: 6810-2020-031)

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